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Single-Photon Depth Reconstruction at Low Signal-Background Ratio Based on Four-Dimensional Attention Mechanism

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Highlights

What are the main findings?

- An end-to-end depth reconstruction network is proposed. Through the pioneering Wavelet-Attention Gating Module and the four-dimensional attention mechanism, it achieves highly efficient extraction of extremely sparse signal photons.
- High-precision depth reconstruction can still be achieved under extreme signal-to-background ratio conditions as low as 1:300. By introducing the non-local total variation loss function, the edge blurring artifacts caused by traditional methods are effectively overcome.

What are the implications of the main findings?

- It provides a robust end-to-end algorithmic framework for all-day long-range single-photon LiDAR, expanding the detection limits of single-photon systems for weak echo signals.
- The breakthrough of this technology provides technical support for practical applications that require early detection and three-dimensional profile recognition of long-range targets.

Abstract

Single-photon Light Detection and Ranging (LiDAR), which is capable of detecting single-photon signals, has developed rapidly in the field of long-range imaging. Due to the long detection range and limited laser power, the accumulated signal photons of single-photon LiDAR are extremely sparse. Meanwhile, the dark current counts, backscattering noise, and background noise of the single-photon detector are significant, resulting in an extremely low signal-background ratio of the detection data. However, existing algorithms struggle to accomplish the depth reconstruction on data with extremely low signal-to-background ratio (SBR). To address the challenges of complex spatiotemporal correlation and feature sparsity in long-range single-photon imaging depth reconstruction, we design a deep reconstruction algorithm based on a classification formulation, specifically tailored for single-echo detection scenarios. We propose a wavelet denoising preprocessing module and a four-dimensional attention module to learn the spatiotemporal correlations of the photon-counting cube data. Sawtooth-arranged dilated convolutions are utilized during the pixel-wise denoising process to extract sparse features, and non-local total variation regularization combined with cross-entropy is introduced as a joint loss function. For depth reconstruction of data with an SBR of 1:100, the root-mean-square error is less than 0.022 m, which is 66.72% lower than that of the best baseline algorithm. It also achieves promising depth reconstruction results on data with an SBR of 1:300.

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Keywords: single-photon LiDAR; depth reconstruction; four-dimensional attention; non-local total variation

1. Introduction

Time-correlated single-photon counting (TCSPC) Light Detection and Ranging (LiDAR) is capable of responding to energy at the single-photon level and is commonly referred to as single-photon LiDAR. The detection range of single-photon LiDAR can reach tens of kilometers, and it is widely applied in fields such as topographic surveying [1], environmental monitoring [2], and military operations [3]. Single-photon LiDAR is capable of detecting weak optical signals and can be utilized in imaging through fog [4], underwater imaging [5], non-line-of-sight imaging [6], and long-range imaging [7]. Single-photon LiDAR employs single-photon avalanche diodes (SPADs) as optical signal acquisition devices to record the time-of-flight (ToF) of echo signal photons from multiple laser pulses. Each pixel of the SPAD independently accumulates photon-counting information to form a histogram, constructing a photon-counting cube that can be used for target depth reconstruction. Affected by backscattering noise, ambient background noise, and dark current counts, the photon-counting cube acquired by the SPAD contains a massive amount of noise counts. In the field of long-range single-photon imaging, Pawlikowska [8] employed time-gating technology, opening the gate only within the expected arrival time of the echo photons. This approach significantly reduced the interference from backscattering and background noise, achieving single-photon imaging at a range of 10 km. Wu [9] developed a common-aperture detection system combining a single-photon laser with mid-wavelength infrared detection, achieving the detection of distant aircraft using a coaxial catadioptric optical module. Li [10] achieved single-photon imaging of a static target at a distance of 201.05 km with 0.44 signal photons per pixel by utilizing a coaxial single-photon LiDAR system equipped with a high-transmittance lens and a high-extinction-ratio transmit/receive module.

In long-range imaging tasks, due to the influence of long distances and limited laser power, the signal photons collected by single-photon LiDAR are extremely sparse. The low ratio of signal photons to background noise photons (signal-to-background ratio, SBR) poses a severe challenge to depth reconstruction algorithms for single-photon LiDAR imaging. Achieving target depth reconstruction from a noisy photon-counting cube is a key research direction. Currently, depth reconstruction algorithms generally fall into two categories: statistical estimation-based algorithms and deep learning-based algorithms.

Statistical estimation-based algorithms first establish a mathematical model for echo photon detection, and then utilize the spatiotemporal correlations of the physical scene to filter out noise through optimization methods such as sparse regularization. Shin [11] proposed a single-photon camera imaging framework that accurately reconstructs depth and reflectivity images from coarse time-resolved data with an average of approximately 1 signal photon per pixel by jointly exploiting the transverse smoothness and longitudinal sparsity of the scene, achieving extremely high photon efficiency. Rapp [12] focused on the separation of signal and noise. Through adaptive range gating and neighborhood data borrowing, they improved the accuracy of depth and reflectivity estimation under conditions with an SBR as low as 0.04, making it applicable to high background noise environments to some extent. Wu [13] proposed an intensity-guided depth estimation method that utilizes the spatiotemporal correlations of the reflected signal and the edge texture information of the intensity image to achieve efficient and

accurate depth reconstruction in long-range LiDAR scenarios with low signal photon returns, while maintaining low computational complexity.

Statistical estimation-based algorithms often perform poorly when processing low SBR data; therefore, many researchers have turned to deep learning methods, achieving significant progress in depth reconstruction under low SBR conditions. Early studies predominantly relied on multimodal data fusion or local priors. For instance, Lindell et al. [14] proposed a deep sensor fusion network that combines the photon arrival times of low-resolution SPAD with high-resolution conventional intensity images; meanwhile, Sun et al. [15] developed the SPADnet framework, fusing RGB images with SPAD data to facilitate monocular depth estimation. However, such multimodal fusion methods impose extremely stringent requirements on cross-modal alignment. Moreover, when the SBR falls below 2:50, intense background noise can easily lead to cross-guidance distortion. To operate under more severe noise environments, researchers have begun to introduce global correlation and edge guidance mechanisms. Peng et al. [16] designed a non-local neural network that captures long-range spatiotemporal correlations in photon data, thereby extending the minimum reconstructable SBR to 1:100. Zhao et al. [17] proposed a method based on full-scale feature integration and intensity edge guidance; their subsequent work [18] further designed a lightweight pyramid network to reduce computational overhead. However, when the SBR drops to 2:100, their depth reconstruction exhibits limitations due to the constrained extraction of sparse features. Yao et al. [19] introduced a pixel-level residual shrinkage network that suppresses intense noise using adaptive optimal thresholds, achieving photon-efficient imaging under a 1:100 SBR condition. To improve the high-spatial-resolution depth reconstruction performance of single-photon LiDAR, Zhang et al. [20] proposed a context-aware spatiotemporal modeling method, enhancing the processing capability for high-spatial-resolution data; however, its depth reconstruction still suffers from limitations when the SBR falls below 2:50. Nevertheless, when facing extreme scenarios in long-range imaging characterized by extreme sparsity and complex spatiotemporal correlations, especially when the SBR further degrades to 1:300 or lower, the aforementioned methods yield limited reconstruction results due to the lack of fine-grained representation of sparse features. Table 1 presents a comparison of the minimum manageable SBRs between the proposed method and other single-photon depth reconstruction methods.

To improve the depth reconstruction performance of long-range single-photon LiDAR, this paper proposes an end-to-end deep neural network that takes the raw SPAD photon-counting cube data as input. To enhance the denoising and filtering effect on background noise photons, a preprocessing module based on wavelet transform is designed. To strengthen the processing capability for extremely low SBR data, a four-dimensional attention module is designed to handle long-range spatial correlations and long-term temporal correlations, focusing on the feature information in sparse data from four dimensions: channel correlation, spatial correlation, temporal correlation, and frequency-domain correlation. Furthermore, to balance the constraint effects between edge features and surface smoothness, a joint loss function combining cross-entropy and non-local total variation regularization is designed. In the depth reconstruction experiments on the Middlebury dataset, the algorithm demonstrates excellent performance in target depth reconstruction for low SBR data, indicating that it can be effectively applied to the field of long-range single-photon imaging.

Table 1. Comparison of the minimum manageable SBR between the proposed method and other single-photon depth reconstruction methods.

Reference	Network Architecture	Core Mechanism	Minimum Manageable Signal-to-Background Ratio (SBR)
Lindell et al. [14]	Deep sensor fusion network	Multi-scale convolution, RGB-SPAD fusion	2:50
Sun et al. [15]	SPADnet	Monocular depth assistance, multimodal fusion	2:50
Zhang et al. [20]	Context-aware spatiotemporal network	Context-aware spatiotemporal modeling	2:50
Zhao et al. [17,18]	Full-scale integration/Pyramid network	Intensity edge guidance, multi-scale fusion	2:100
Peng et al. [16]	Non-local neural network	Long-range spatiotemporal correlation capture	1:100
Yao et al. [19]	Residual shrinkage network	Pixel-level adaptive threshold shrinkage	1:100
Ours	Asymmetric Multi-scale Attention-Guided Denoising Network	Attention guidance, wavelet denoising preprocessing	1:300

2. Materials and Methods

2.1. Long-Range Single-Photon Imaging Model

The long-range single-photon LiDAR system consists of an optical lens, a laser, a SPAD, and a post-processing module, as shown in Figure 1; owing to the extremely low SBR, the shape contours cannot be intuitively discerned within the photon-counting cube. In long-range detection scenarios, the laser operates at high power and, after collimation, emits laser pulses toward the target. The large-aperture optical lens is beneficial for capturing the returning signal photons, and the SPAD acquires the echo signals. The post-processing module utilizes TCSPC technology to construct a photon-counting cube, which is then processed by algorithms to reconstruct the target depth map.

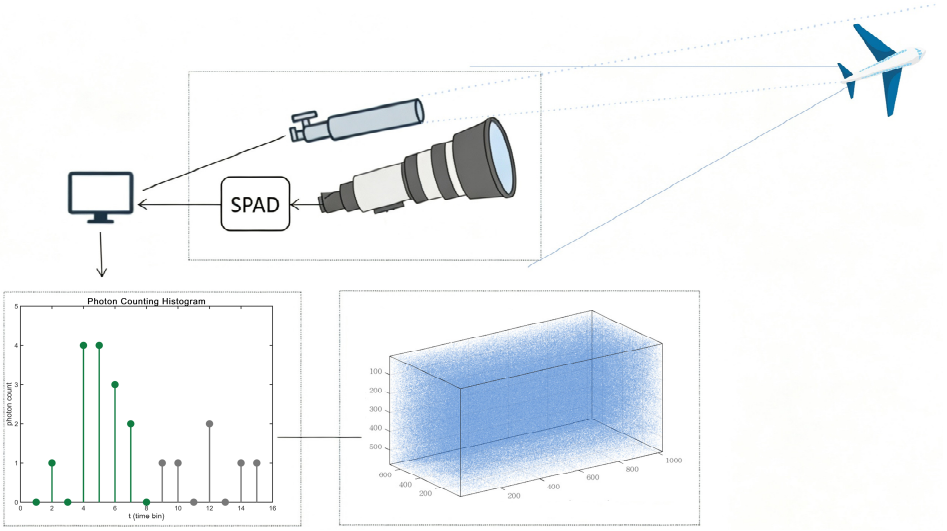


Figure 1. Schematic diagram of a long-range single-photon LiDAR system.

To ensure the accuracy of the single-photon imaging process, two constraints are introduced before establishing the imaging model of the long-range single-photon LiDAR system. To avoid periodic overlap in the distances calculated by ToF, which causes range

ambiguity, the pulse repetition period T_r emitted by the laser needs to be sufficiently long to cover the maximum detection range $R_{\max}$ of the target, where the speed of light in a vacuum is c. Under this constraint, all echoes from the previous pulse have returned before the next laser pulse is emitted, ensuring that every received photon corresponds uniquely to a single laser pulse.

$$T_r \geq \frac{2R_{\max}}{c}, \tag{1}$$

SPADs exhibit a dead time, during which they cannot detect new photons for a certain period after a photon has been detected. Under high-flux conditions, this pile-up effect can cause signal distortion. It is assumed that the average photon detection probability per pulse repetition period is less than 5% to neglect the photon pile-up effect [21–23].

2.2. Depth Reconstruction Strategy

Specifically tailored for single-echo detection scenarios, this study formulates the single-photon imaging depth reconstruction task as a pixel-wise classification task in the temporal domain. It classifies the time bins associated with signal photons versus noise photons across all time bins, achieving depth reconstruction in an end-to-end manner. Regarding the classification strategy, probabilistic prediction of the time bins is performed during the feature extraction stage. In the depth estimation stage, to overcome the quantization error introduced by the time bin width, the SoftArgmax approach is adopted to calculate the expected value.

For a spatial pixel location (i, j), the neural network f_θ maps the noise histogram to the probability along the time axis, converting it into a normalized probability distribution $p_{i,j}$. The output value of the network at the k time bin is $z_{i,j}(k)$, and the posterior probability of the signal photon arriving at the k time bin is $p_{i,j}(k)$.

$$p_{i,j}(k) = \frac{\exp\left(z_{i,j}(k)\right)}{\sum_{m=1}^{T} \exp\left(z_{i,j}(m)\right)}, \tag{2}$$

To obtain high-precision depth estimation, rather than directly selecting the index with the maximum probability, the mathematical expectation of the index probability distribution is calculated. The predicted time bin index is $\hat{\tau}(i, j)$.

$$\hat{\tau}(i, j) = \sum_{k=1}^{T} k \cdot p_{i,j}(k), \tag{3}$$

For single-echo detection scenarios, classification models offer distinct advantages over regression models. Since classification models focus on identifying the time bin with the highest probability, they are less susceptible to the effects of random noise. Furthermore, this classification approach reduces the processing dimensionality of the depth reconstruction task, thereby enhancing both reconstruction accuracy and computational efficiency.

2.3. Asymmetric Multi-Scale Attention-Guided Denoising Network Architecture

To address the challenge of extremely sparse signal photons in long-range single-photon LiDAR depth reconstruction, this study proposes an Asymmetric Multi-scale Attention-Guided Denoising Network (AMAD-Net), as shown in Figure 2. Considering the characteristics of photon-counting cube data, AMAD-Net employs an asymmetric structure comprising a cascaded downsampling encoder and a transposed convolutional decoder, thereby constructing a depth reconstruction pipeline that includes feature encoding, depth feature refinement, and pixel-wise classification decoding.

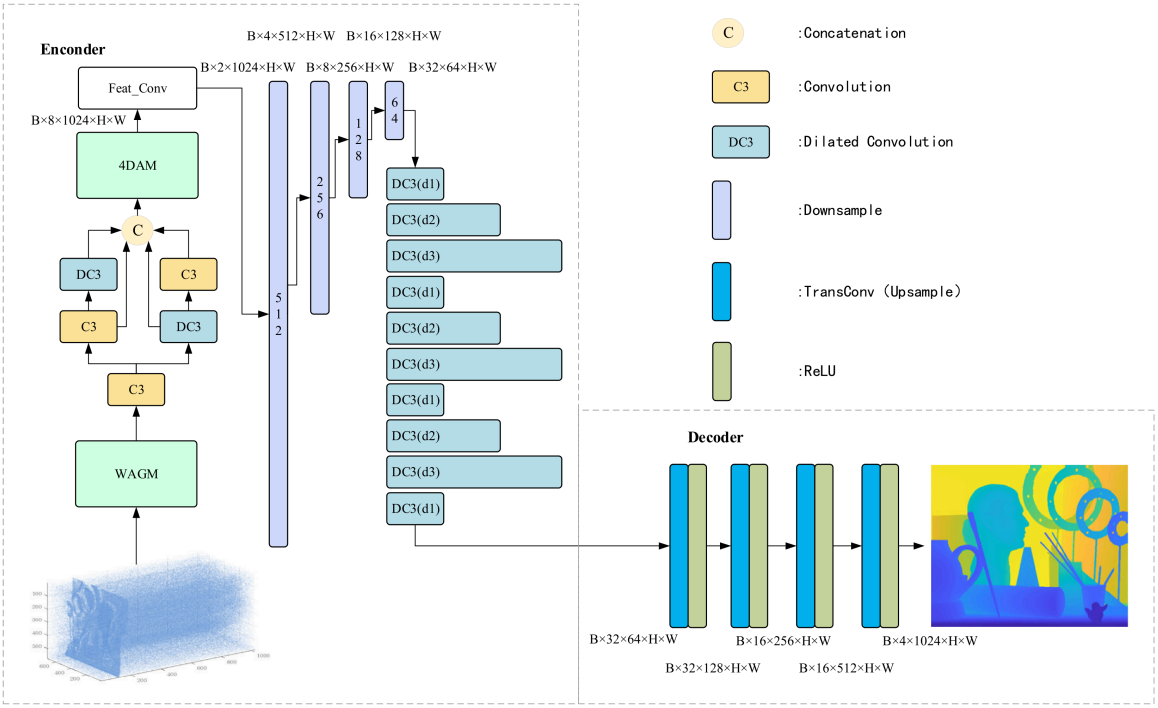


Figure 2. Architecture of the AMAD-Net.

In the feature encoding stage, the feature extraction module proposed by Peng [16] is utilized. This module alternately employs standard three-dimensional (3D) convolutions and dilated 3D convolutions on two parallel branches to alternately capture long-range correlations and local detailed features, achieving the fusion of features from different receptive fields through a cross-input mechanism. To finely capture the spatiotemporal correlations in the data, the input time dimension is progressively compressed by a factor of 16 through cascaded downsampling, and the number of feature channels is expanded to 32. This approach enlarges the receptive field of the network while improving the algorithm's efficiency.

In the depth feature refinement stage, the encoded high-dimensional features enter the feature refinement module enhanced by four-dimensional attention. Simultaneously, a sawtooth-arranged dilated convolution strategy is introduced to construct a multi-scale receptive field within the feature space. A collaborative approach combining global average pooling and max pooling is employed to dynamically and adaptively generate a soft threshold based on the absolute intensity of the signal. When the maximum value is significantly greater than the average value, the threshold is automatically relaxed to preserve transient signals; conversely, when the maximum value is close to the average value, the threshold is automatically tightened to filter out noise.

In the pixel-wise classification decoding stage, the reconstruction module takes the refined features as input, utilizes 3D transposed convolutional layers to fuse multi-channel feature information, and progressively restores the time-dimension resolution. The Soft-Argmax method is adopted to calculate the expected value of the index probability distribution pixel by pixel, where the time bin with the highest probability corresponds to the signal photon.

2.4. Wavelet Denoising Preprocessing Module

The Wavelet-Attention Gating Module (WAGM) suppresses the noise photon counts, as shown in Figure 3. In long-range single-photon imaging, photon arrival events exhibit discrete and concentrated transient singularities in the histogram. According to Mallat's

wavelet multi-resolution analysis theory [24], the local singularity of a signal determines the magnitude of its wavelet coefficients.

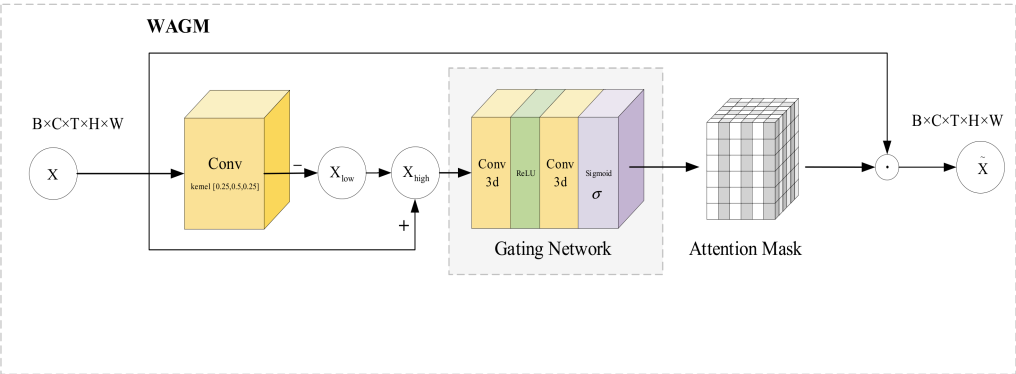


Figure 3. Architecture of the WAGM.

To intuitively illustrate this physical property, Figure 4 presents the noisy photon count histogram alongside its high-frequency coefficient distribution after the stationary wavelet transform (SWT). As shown in Figure 4a, the raw photon pulses are extremely sparse along the temporal axis. In contrast, Figure 4b shows that the SWT precisely captures the transient edges of each photon arrival event, manifesting as sharp peaks with large coefficients in the high-frequency subbands, while the regions without photon arrivals decay to near-zero values. This experimental phenomenon demonstrates that high-frequency wavelet features can effectively highlight photon pulses at the feature level.

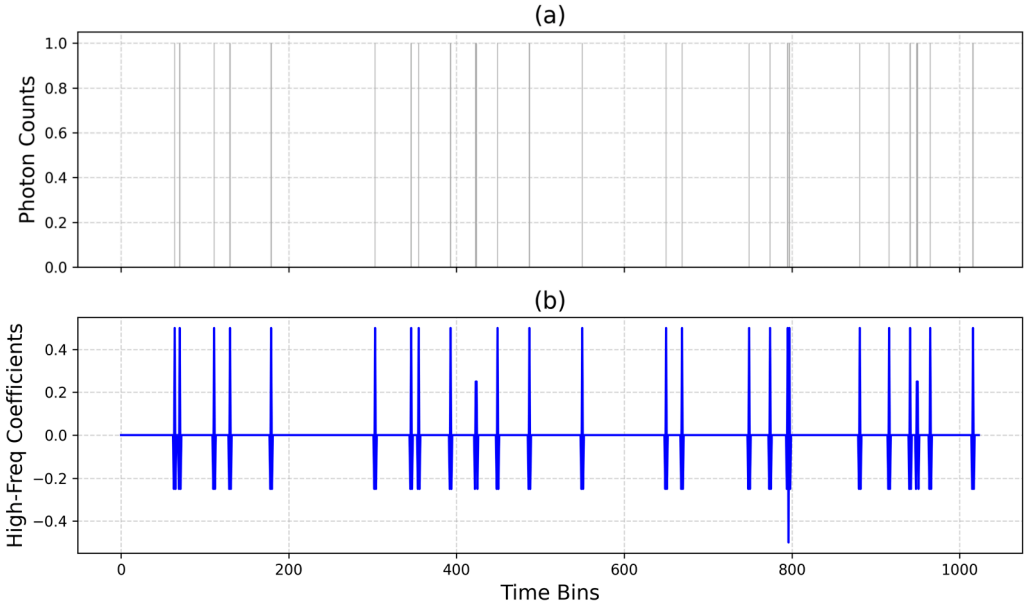


Figure 4. Schematic diagram of photon pulse extraction via wavelet transform: (**a**) raw noisy photon count histogram; (**b**) wavelet high-frequency coefficients.

Based on the aforementioned physical characteristics and experimental evidence, we designed the WAGM as a lightweight, end-to-end differentiable module placed at the very front of the deep single-photon reconstruction network. It operates through three consecutive steps: high-frequency feature extraction, attention masking, and multiplicative denoising. First, a phase-shift-free stationary wavelet transform is utilized to extract high-frequency detail components, converting all transient photon events into salient features with large coefficients. Second, to distinguish signal photons from random Poisson noise

within these discrete events, we employ an asymptotic gating strategy. Since signal photons exhibit strong spatiotemporal correlation in local neighborhoods, whereas Poisson noise is spatiotemporally independent and uniformly distributed, this strategy generates pixel-wise attention masks by learning spatiotemporal features. This soft-masking approach gradually suppresses the weights of isolated background noise regions. Unlike conventional soft-thresholding denoising methods that may truncate weak signals, the WAGM integrates wavelet physical priors with a spatiotemporal attention mechanism. By adaptively suppressing the weights of noisy regions, it preserves the integrity of the signal photon regions to the maximum extent.

2.5. Four-Dimensional Attention Module

The four-dimensional attention module (4DAM) is positioned after the feature extraction module, as shown in Figure 5. The definition of "four-dimensional" (4D) is explicitly clarified. Although the raw data collected by the SPAD sensor is a three-dimensional tensor consisting of spatial height, spatial width, and range, the front-end convolutional operations map this data into a high-dimensional abstract feature space, thereby introducing the feature channel dimension of the convolutional neural network. Consequently, the four-dimensional (4D) attention module establishes a hybrid attention mechanism integrating channel, temporal, frequency-domain (transformed from temporal features), and spatial correlations. Regarding the overall operational mechanism, the 4DAM adopts an architecture of "parallel shunting, independent perception, cross-domain enhancement, and gated fusion." Specifically, the input features are first split in parallel into three independent branches targeting channel, global temporal, and local temporal correlations, respectively, to extract feature descriptors. Subsequently, a spatial-frequency domain perception module is cascaded at the end of each branch to perform spatial structural enhancement. Finally, through an adaptive gating mechanism, the multi-dimensional attention responses are fused and applied to modulate the original input features. This mechanism comprehensively tackles the extreme sparsity of signal photons induced by long distances, thereby enabling the network to extract more effective and robust features under low SBR conditions.

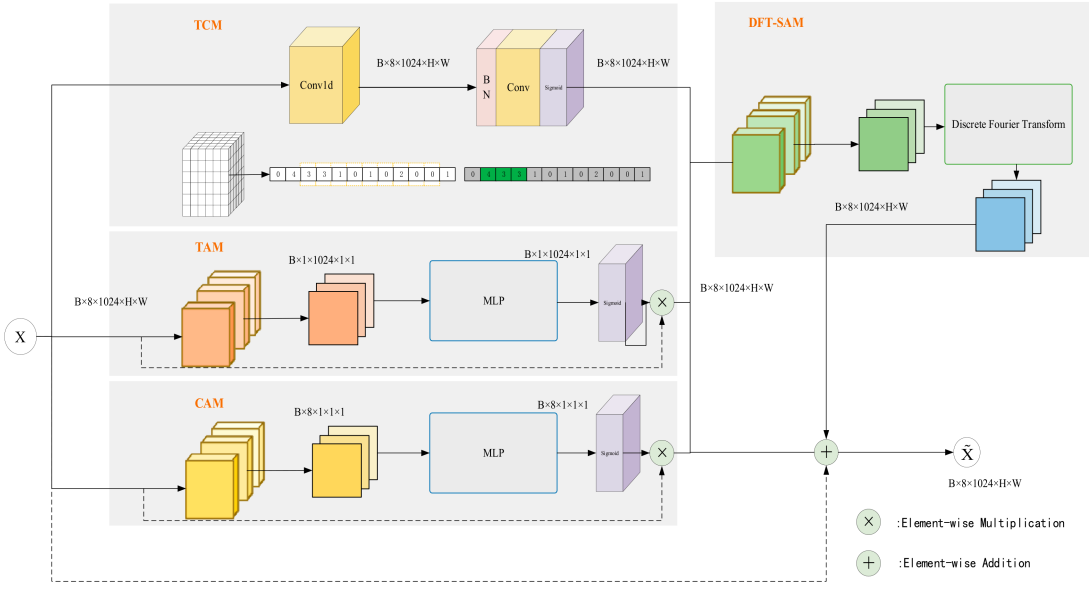


Figure 5. Architecture of the 4DAM module.

The channel correlation is implemented through the Channel Attention Module (CAM) branch of the CBAM [25]. It should be particularly emphasized that since single-photon

LiDAR only detects time-of-flight information, the "channel" here refers to the high-order semantic feature map dimensions generated during the feature mapping process of the convolutional neural network. Each feature channel represents a distinct feature subspace learned by the network. In operation, the CAM first performs global average pooling and global max pooling on the input feature tensor to compress complex spatial and temporal information into channel-wise background statistics and salient response features, respectively. Subsequently, these two aggregated features are fed into a shared multi-layer perceptron (MLP) to learn the complex non-linear dependencies among different channels. Finally, the network generates normalized channel weight coefficients to modulate the input features, thereby effectively extracting channel-wise correlated features that contain high-confidence signal photons while dynamically suppressing noise-dominated channels.

The extraction of temporal correlation features is divided into micro and macro levels, which are implemented in parallel via the proposed Temporal Correlation Module (TCM) and TAM [26], respectively. At the micro level, TCM focuses on capturing local waveform correlations along the temporal axis. Considering that real laser echoes exhibit specific pulse distributions, TCM utilizes 1D large-kernel convolutions to perform sliding-window detection along the temporal dimension. This operation specifically matches the outer envelope shape of signal photon counts while preserving spatial information. Leveraging the continuous aggregation characteristics of signal photon-counting histograms, this mechanism can accurately distinguish true signal count envelopes from randomly isolated noise photon-counting spikes. Compared with TAM, the local denoising and envelope extraction performed by TCM focus more on the microscopic morphology of signal photons, thereby serving as an effective complement to TAM. If the raw, highly noisy histogram data were processed solely by the global module, feature extraction could easily be misled by randomly distributed extreme noise peaks. At the macro level, TAM is responsible for evaluating global temporal correlations. Its operational mechanism involves establishing cross-dimensional relationships between the temporal axis and the spatial and channel dimensions through dimension rotation and permutation operations. Subsequently, pooling is performed over the spatial dimensions (H and W) and channels to obtain the average photon counts at each discrete time bin from a global perspective. By sacrificing channel and local spatial variations, TAM determines the temporal segments where the photon counts of the entire scene are concentrated from a macroscopic perspective, utilizing global information to roughly yet robustly identify the region of interest along the temporal axis.

To address frequency and spatial correlations, we propose the Discrete Fourier Transform-based Spatial Attention Module, which draws inspiration from the SAM branch in CBAM [25] and is cascaded after each of the aforementioned parallel branches as a spatial structural enhancer. Typically, the photon-counting cube captured by the SPAD is highly sparse, where extremely scarce signal photon counts are deeply buried in abundant noise photon counts. Directly extracting spatial features through pooling in the temporal domain not only risks losing these sparse, weak signals but is also highly susceptible to being misled by noise photons. In contrast, physical characteristics differ significantly in the frequency domain: signal photon counts generally manifest as energy-concentrated low- and mid-frequency components, whereas noise photon counts exhibit a uniform distribution across the entire spectrum or represent high-frequency energy perturbations. Therefore, the Discrete Fourier Transform-based Spatial Attention Module adopts an innovative cross-domain perceptual architecture consisting sequentially of a frequency-domain feature extraction stage and a spatial attention generation stage. In the frequency-domain feature extraction stage, the module performs a Discrete Fourier Transform on the temporal count sequence of each pixel, transforming the extremely sparse temporal histograms into dense frequency-domain energy distribution spectra to achieve feature de-sparsification.

Subsequently, in the spatial attention generation stage, the module utilizes the high-quality features extracted from the frequency domain to identify spatial regions of interest. Specifically, it aggregates the maximum and average statistics of the frequency spectrum along the spatial dimensions to generate a comprehensive spatial-frequency descriptor. A two-dimensional convolutional layer is then employed to learn the spectral energy variation patterns among adjacent spatial pixels, precisely mapping the frequency-domain energy differences to attention weight distributions in the geometric space. Finally, the resulting attention map, which integrates multi-dimensional perception, guides the network to focus on high-confidence signal regions.

2.6. Pixel-Wise Denoising Module

The pixel-wise denoising module acts as the decoder of the network. It assigns an appropriate denoising threshold to each pixel, setting feature points smaller than the threshold to zero while retaining only those larger than the threshold, thereby achieving feature-level denoising. In standard convolutions, the elements processed by the convolution kernel are arranged tightly together. In dilated convolutions, the convolution kernel is modified by introducing empty spaces between its elements. Without increasing the number of parameters or computational complexity, setting dilated regions within the convolution kernel effectively expands its coverage area over time, which is beneficial for enlarging the receptive field. To preserve the spatial feature information of sparse signal photons, we employ 3D dilated convolutions in the feature extraction module. This approach does not use strided convolutions to increase the receptive field; instead, it enlarges the receptive field by increasing the dilation rate without reducing the resolution of the feature maps. This allows it to capture more neighborhood information while retaining the original spatial resolution, thereby effectively extracting spatial features surrounding the target region. To avoid the loss of local information caused by discontinuous convolution kernels, a sawtooth-arranged dilation rate pattern is designed. The sawtooth-arranged dilation rates ensure that different features can be fully fused, taking into account both high-frequency details and low-frequency information, which is beneficial for improving the detection capability of weak signals.

To more accurately separate effective signals from background noise, this module introduces an improved dual-statistic threshold generation mechanism. Traditional threshold generation often relies solely on Global Average Pooling (GAP) to estimate the background noise level, which tends to ignore sparse, strong signal peaks. In our design, we calculate the absolute value of the features along the temporal dimension, extracting the GAP feature to represent the overall ambient background noise level, alongside the Global Max Pooling (GMP) feature to capture the strongest photon signals or abrupt noise spikes. The shape feature of the original waveform, normalized by its maximum value, is concatenated along the channel dimension with these two absolute intensity statistics. This structural design enables the threshold generation network to simultaneously perceive the explicit waveform structure and the absolute intensity of the signal. It dynamically outputs a more precise scaling factor for each pixel, and the final pixel-wise threshold is obtained by multiplying the base threshold by this scaling factor.

Regarding the choice of activation functions, considering that single-photon detection data is inherently sparse and weak, the traditional ReLU activation function hard-truncates all negative values to zero. This behavior easily leads to the loss of subtle effective features in deep networks and the dead neuron problem. This module employs the Parametric Rectified Linear Unit (PReLU). By introducing a learnable slope parameter on the negative half-axis, PReLU allows subtle negative gradients to pass through during the non-linear mapping process. This modification strictly protects the integrity of weak photon signals

and significantly enhances the stability and representational capacity of the network during deep feature extraction. A soft thresholding operation is applied using the generated dynamic thresholds, shrinking the absolute values of the features and adding the processed residuals back to the original input, thereby accomplishing high-fidelity pixel-wise denoising and depth reconstruction.

2.7. Loss Function

Since the network does not directly regress depth values but predicts the probability distribution over time bins to distinguish signal photons from background noise photons, we employ a cross-entropy-based loss function. By simplifying long-range single-photon imaging to a unimodal depth reconstruction scenario, the cross-entropy loss is used to constrain the shape of the probability distribution, making it continuously approximate the true probability distribution. This forces the network to concentrate the probability distribution around the true signal bin, ensuring accurate peak localization performance.

For a spatial pixel location (i, j), the ground-truth depth map is $D_{i,j}$, and the continuous $D_{i,j}$ is mapped to a discrete time bin index $I_{i,j}$. The size of the time bin is Δ, and the total number of time bins is T. Through one-hot encoding, $P \in \mathbb{R}^{M \times N \times T}$ is obtained from $I_{i,j}$ as the classification label. The true distribution label is $p_{i,j}$, and the predicted probability distribution label is $\hat{p}_{i,j}$. The true distribution of the k time bin is $p_{i,j,k}$, and the predicted probability of the k time bin is $\hat{p}_{i,j,k}$. The cross-entropy loss function between the output and the true label is $\mathcal{L}_{CE}$.

$$I_{i,j} = \frac{2D_{i,j}}{\Delta \cdot c}, \tag{4}$$

$$\mathcal{L}_{CE}(p_{i,j}, \hat{p}_{i,j}) = -\sum_{k=1}^{T} p_{i,j,k} \log \hat{p}_{i,j,k}, \tag{5}$$

In previous studies, total variation (TV) was often used as a regularization term. TV ensures smoothness in local regions by minimizing the gradients between adjacent pixels. However, experiments have revealed that the local smoothing introduced by TV destroys high-frequency depth details. In the context of long-range single-photon imaging depth reconstruction, the data has an extremely low signal-to-background ratio, causing local gradients to become unstable and susceptible to noise artifacts. To avoid these issues, we introduce the non-local total variation (NLTV) [27] loss. NLTV fully utilizes the non-local self-similarity of the target depth map; the effective structural edges of the target scene have a high probability of reappearing within a larger search neighborhood, whereas noise, by contrast, lacks such non-local self-similarity.

NLTV is based on a search window W_p centered on pixel p, with the image domain being Ω, and a normalization factor Z based on the window size. The structural similarity between a patch P_p centered on pixel p and a patch P_q centered on pixel q is represented by the weight $w_{p,q}$. The mean square error between the two patches is $MSE(P_p, P_q)$, the decay parameter controlling the filtering strength is h, the NLTV loss function is $\mathcal{L}_{NLTV}$, and the reconstructed depth is $\hat{D}$.

$$w_{p,q} = \exp\left(-\frac{MSE(P_p, P_q)}{h^2}\right), \tag{6}$$

$$\mathcal{L}_{NLTV}(\hat{D}) = \frac{1}{Z}\sum_{p \in \Omega}\sum_{q \in W_p} w_{p,q}|\hat{D}_p - \hat{D}_q|, \tag{7}$$

When processing smooth regions, the weight $w_{p,q}$ is relatively high, enabling the network to filter out noise by utilizing information from multiple similar neighborhoods.

When processing edge regions with depth discontinuities, the weight $w_{p,q}$ decreases significantly due to the low patch matching degree, thereby avoiding blurring artifacts while preserving sharp edge information.

2.8. Single-Photon Dataset

Consistent with recent single-photon imaging depth reconstruction algorithms [14–20], the dataset utilizes photon-counting data simulated based on the NYUv2 dataset [28] to mimic the acquisition by single-photon LiDAR SPADs. In the simulation algorithm, the Full Width At Half Maximum of the laser beam waveform is 400 ps, and the time resolution is 80 ps. Similarly, 9 conventional SBR levels (2:2, 2:10, 2:50, 5:2, 5:10, 5:50, 10:2, 10:10, and 10:50) and 3 lower SBR levels (1:100, 2:100, and 3:100) are defined. The time dimension of the data contains 1024 time bins, and the spatial dimensions are 64 in height and 64 in width. The training set comprises 13,244 data samples, and the validation set contains 2753 data samples.

To validate the generalization capability of the algorithm, data consistent with the aforementioned conventional and lower SBR levels were established based on the photon-counting data acquired by the single-photon LiDAR SPAD across 8 scenes from the Middlebury dataset [29]: Art, Books, Bowling1, Dolls, Laundry, Moebius, Plastic, and Reindeer.

2.9. Experimental Setup

As shown in Table 2, the experimental platform was a desktop computer equipped with an Intel Core i5-14600F @ 3.5GHz CPU and an NVIDIA GeForce RTX 4080 SUPER GPU, running the Ubuntu 24.04.2 LTS operating system. The simulation algorithm for the simulated scene dataset was executed in the MATLAB R2024a environment. The Conda environment for the deep learning algorithms utilized Python 3.12.7, PyTorch 2.5.1, and CUDA 12.4, with program debugging performed in PyCharm 2025.3.4. The NYUv2 dataset was used to simulate the single-photon LiDAR detection results, and the data was randomly cropped to $32 \times 32 \times 1024$. During the training of the AMAD-Net algorithm, the batch size of the training data was set to 8, and the model was trained for a total of 29 epochs. The Adam optimizer was employed with an initial learning rate of 1×10^{-3}, an exponential decay rate for the first-moment estimates of 0.9, and an exponential decay rate for the second-moment estimates of 0.999. The learning rate was decayed by a factor of 0.6 every 3 epochs. Regarding the loss function, the coefficient for Cross-Entropy (CE) was set to 1, and the coefficient for NLTV was set to 1×10^{-6}. For NLTV, the search window size was set to 5, the patch size for calculating similarity was set to 3, and the noise weight coefficient was set to 0.2.

Table 2. Implementation details of the experimental platform.

Item	Specification
CPU	Inter(R) Core™ i5-14600F@3.5GHz
GPU	NVIDIA GeForce RTX 4080 SUPER
GPU Memory	16G
Operating System	Ubuntu 24.04.2 LTS
Deep Learning Framework	Pytorch 2.5.1 + CUDA 12.4

2.10. Evaluation Metrics

To effectively evaluate the depth reconstruction performance of the photon-counting data acquired by the single-photon LiDAR SPAD from multiple perspectives, the Root Mean Square Error (RMSE) was used to assess the overall error level. The accuracy metric was utilized to evaluate the precision of depth reconstruction within different error ranges. The Structural Similarity Index (SSIM) was used to evaluate structural similarity before

and after reconstruction, while the Peak Signal-to-Noise Ratio (PSNR) was used to assess the similarity in details before and after reconstruction. Additionally, the Absolute Relative Error (AbsRel) and Squared Relative Difference (SqRel) were employed to evaluate the relative differences before and after reconstruction.

RMSE: The root mean square error between the ground-truth depth map D and the reconstructed depth map $\hat{D}$, where M is the total pixel height of the depth map and N is the total pixel width of the depth map.

$$RMSE(\hat{D}, D) = \sqrt{\frac{1}{MN}\sum_{i}^{M}\sum_{j}^{N}(\hat{D}_{i,j} - D_{i,j})^2}, \tag{8}$$

accuracy: The ratio of the number of pixels with an error less than the threshold δ to the total number of pixels in the ground-truth depth map D and the reconstructed depth map $\hat{D}$, where M is the total pixel height of the depth map and N is the total pixel width of the depth map.

$$accuracy = \text{pixels where } \left[\frac{1}{MN}\sum_{i}^{M}\sum_{j}^{N}\max\left(\frac{\hat{D}_{i,j}}{D_{i,j}}, \frac{D_{i,j}}{\hat{D}_{i,j}}\right) \leq \delta\right], \tag{9}$$

SSIM: The structural similarity index between the ground-truth depth map D and the reconstructed depth map $\hat{D}$, where $\hat{\sigma}$ and $\hat{\mu}$ are the standard deviation and mean of the ground-truth depth map, σ and μ are the standard deviation and mean of the reconstructed depth map, $\widetilde{\sigma}$ is the standard deviation of the pixel differences between the depth maps before and after reconstruction, and C_1 and C_2 are constants to promote computational stability.

$$SSIM(\hat{D}, D) = \frac{1}{n}\sum_{i=1}^{n}\frac{(2\hat{\mu}_i\mu_i + C_1)(2\widetilde{\sigma}_i + C_2)}{\left[(\hat{\mu}_i)^2 + (\mu_i)^2 + C_1\right]\left[(\hat{\sigma}_i)^2 + (\sigma_i)^2 + C_2\right]}, \tag{10}$$

PSNR: The peak signal-to-noise ratio between the ground-truth depth map D and the reconstructed depth map $\hat{D}$.

$$PSNR(\hat{D}, D) = 20\log_{10}\frac{\max(D)}{\sqrt{\frac{1}{MN}\sum_{i}^{M}\sum_{j}^{N}(\hat{D}_{i,j} - D_{i,j})^2}}, \tag{11}$$

AbsRel: The absolute relative difference between the ground-truth depth map D and the reconstructed depth map $\hat{D}$, where M is the total pixel height of the depth map and N is the total pixel width of the depth map.

$$\text{AbsRel}(\hat{D}, D) = \frac{1}{MN}\sum_{i}^{M}\sum_{j}^{N}\frac{\left|D_{i,j} - \hat{D}_{i,j}\right|}{D_{i,j}}, \tag{12}$$

SqRel: The squared relative difference between the ground-truth depth map D and the reconstructed depth map $\hat{D}$, where M is the total pixel height of the depth map and N is the total pixel width of the depth map.

$$\text{SqRel}(\hat{D}, D) = \frac{1}{MN}\sum_{i}^{M}\sum_{j}^{N}\frac{\left(D_{i,j} - \hat{D}_{i,j}\right)^2}{D_{i,j}}, \tag{13}$$

3. Results

3.1. Experimental Results on the Middlebury Dataset

3.1.1. Metric Evaluation of Multiple Algorithms

The experimental results of the proposed AMAD-Net algorithm were compared with those of the LM algorithm, Shin algorithm, Rapp algorithm, Lindell algorithm, Peng algorithm, and Yao algorithm on the Middlebury dataset.

SBR data at levels of 2:50, 1:100, 2:100, and 3:100 were used as representatives of lower SBR levels to quantitatively evaluate the performance of the AMAD-Net algorithm and other algorithms, the average values of the evaluation parameters across the 8 scenes were statistically analyzed, as shown in Tables 3–8.

Table 3. Evaluation of multiple algorithms using the RMSE (m) metric.

Types of SBR	LM	Shin	Rapp	Lindell	Peng	Yao	Proposed
1:100	6.8415	5.6974	0.9464	0.7467	0.1691	0.0658	0.0219
2:50	5.8182	5.4753	0.3122	0.1115	0.0192	0.0150	0.0133
2:100	6.1822	5.6238	0.4807	0.1834	0.0309	0.0229	0.0146
3:100	5.5156	5.5482	0.3100	0.1330	0.0153	0.0130	0.0124

Table 4. Evaluation of multiple algorithms using the PSNR (dB) metric.

Types of SBR	LM	Shin	Rapp	Lindell	Peng	Yao	Proposed
1:100	−8.2548	−6.6631	10.3536	12.1771	24.6948	35.3130	41.4678
2:50	−6.8476	−6.3176	22.8665	27.8102	42.4629	44.5844	45.6074
2:100	−7.3751	−6.5502	17.6460	25.1263	39.9132	42.5696	44.7533
3:100	−6.3827	−6.4325	23.7516	26.6103	44.3260	45.5492	46.1379

Table 5. Evaluation of multiple algorithms using the accuracy metric.

δ	Types of SBR	LM	Shin	Rapp	Lindell	Peng	Yao	Proposed
1.01	1:100	7.52%	7.00%	62.63%	18.08%	71.18%	82.64%	91.39%
	2:50	27.70%	4.63%	80.54%	16.05%	94.36%	95.19%	97.02%
	2:100	21.30%	6.61%	76.32%	22.23%	91.61%	93.65%	96.28%
	3:100	34.44%	5.71%	82.18%	18.03%	95.75%	96.01%	97.43%
1.02	1:100	9.8%	13.64%	75.17%	65.28%	86.87%	94.96%	97.61%
	2:50	33.99%	9.29%	90.12%	61.22%	98.53%	99.12%	99.27%
	2:100	25.96%	13.11%	86.01%	69.42%	97.90%	98.79%	99.07%
	3:100	40.56%	11.35%	91.74%	64.26%	98.94%	99.35%	99.45%
1.03	1:100	10.37%	19.93%	81.19%	84.03%	90.60%	96.99%	98.56%
	2:50	34.97%	14.01%	93.62%	84.12%	99.16%	99.52%	99.57%
	2:100	26.73%	19.28%	90.05%	86.96%	98.80%	99.32%	99.44%
	3:100	41.36%	16.65%	94.66%	85.34%	99.39%	99.64%	99.66%

Table 6. Evaluation of multiple algorithms using the SSIM metric.

Types of SBR	LM	Shin	Rapp	Lindell	Peng	Yao	Proposed
1:100	0.0001	0.2800	0.8553	0.7553	0.8155	0.9178	0.9556
2:50	0.0005	0.2367	0.9220	0.8539	0.9548	0.9742	0.9799
2:100	0.0003	0.2686	0.9076	0.8453	0.9388	0.9675	0.9762
3:100	0.0027	0.2628	0.9318	0.8495	0.9618	0.9780	0.9823

Table 7. Evaluation of multiple algorithms using the AbsRel metric.

Types of SBR	LM	Shin	Rapp	Lindell	Peng	Yao	Proposed
1:100	3.1913	3.3962	0.1202	0.0747	0.0260	0.0100	0.0054
2:50	2.2888	3.2606	0.0247	0.0289	0.0043	0.0045	0.0037
2:100	2.5891	3.3515	0.0445	0.0299	0.0054	0.0050	0.0040
3:100	2.0523	3.3054	0.0208	0.0290	0.0038	0.0043	0.0036

Table 8. Evaluation of multiple algorithms using the SqRel metric.

Types of SBR	LM	Shin	Rapp	Lindell	Peng	Yao	Proposed
1:100	27.8890	19.5179	0.6771	0.3807	0.0232	0.0059	0.0003
2:50	19.9672	18.0081	0.1064	0.0065	0.0002	0.0001	0.0001
2:100	22.6036	19.0097	0.2349	0.0275	0.0009	0.0005	0.0001
3:100	17.8767	18.4965	0.1064	0.0102	0.0001	0.0001	0.0001

3.1.2. Depth Reconstruction Results of Multiple Algorithms

Taking the Art scene from the Middlebury dataset with an SBR of 1:100 as an example, the depth reconstruction results of different algorithms are demonstrated, as shown in Figure 6.

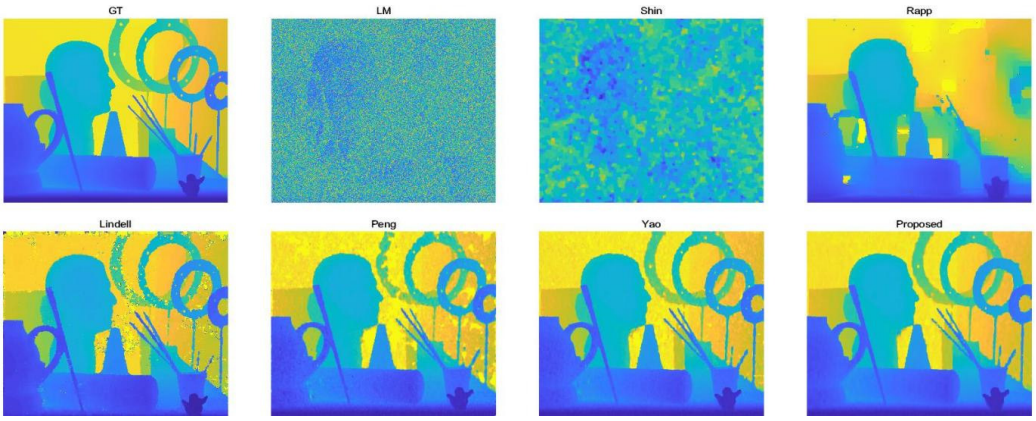


Figure 6. Depth reconstruction performance of different algorithms for the Art scene at SBR = 1:100.

To further evaluate the reconstruction precision of each algorithm quantitatively and visually, Figure 7 presents the corresponding absolute depth error maps for the "Art" scene at an SBR of 1:100. In these error maps, cooler colors represent lower reconstruction errors, whereas warmer colors indicate larger deviations from the ground-truth depth. The statistical estimation-based methods, such as LM and Shin, exhibit widespread, high-magnitude errors across the entire scene due to their vulnerability to intense background noise. Although deep learning baselines such as Lindell et al., Peng et al., and Yao et al. significantly reduce errors on flat surfaces, they still suffer from noticeable deviations at complex structural boundaries and depth discontinuities. In contrast, the absolute error map of our proposed AMAD-Net is almost entirely dominated by dark blue, demonstrating near-zero reconstruction errors across both smooth regions and fine edges. This visual comparison further confirms the superior denoising and structure-preserving capabilities of our method under extremely low SBR conditions.

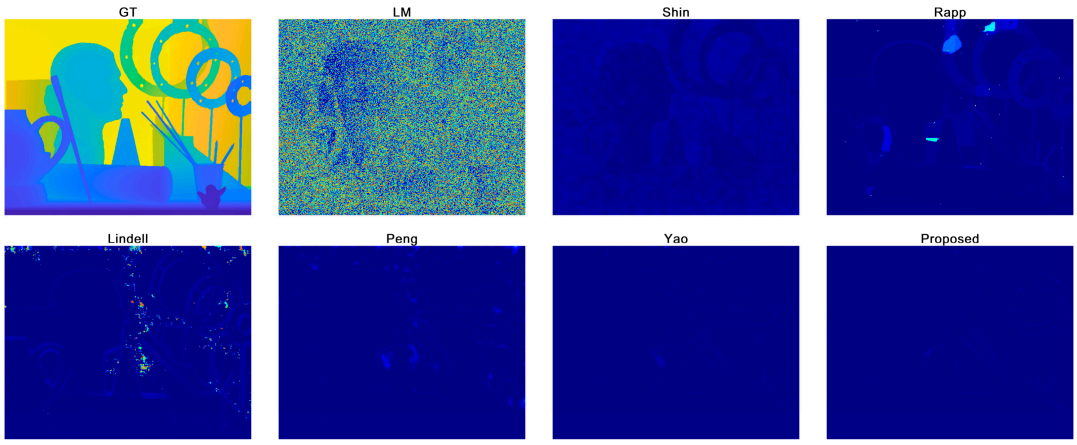


Figure 7. Absolute depth error maps of different algorithms for the Art scene at SBR = 1:100.

3.1.3. Metric Evaluation of Depth Reconstruction for the Proposed Algorithm on Extreme Low SBR Data

Depth reconstruction was performed on 9 types of extremely low SBR data (3:200, 2:200, 1:200, 3:300, 2:300, 1:300, 3:500, 2:500, and 1:500), and the average values of the evaluation parameters across the 8 scenes were statistically analyzed, as shown in Table 9.

Table 9. Evaluation of Depth Reconstruction Performance on Extremely Low SBR Data.

Types of SBR	RMSE(m)	PSNR(dB)	Accuracy (δ = 1.01)	Accuracy (δ = 1.02)	Accuracy (δ = 1.03)	SSIM	AbsRel	SqRel
1:200	0.0512	36.5166	85.31%	95.65%	97.29%	0.9311	0.0083	0.0030
1:300	0.1131	30.6315	77.05%	91.82%	94.64%	0.8900	0.0162	0.0173
1:500	0.5416	13.5214	37.84%	52.97%	59.52%	0.4642	0.1464	0.1973
2:200	0.0186	42.9769	94.72%	98.67%	99.17%	0.9692	0.0046	0.0002
2:300	0.0200	42.3083	92.60%	98.32%	98.97%	0.9629	0.0052	0.0003
2:500	0.0983	32.3389	85.67%	95.25%	96.52%	0.9068	0.0142	0.0118
3:200	0.0140	45.1609	96.55%	99.25%	99.54%	0.9786	0.0040	0.0001
3:300	0.0156	44.2103	95.52%	99.04%	99.41%	0.9749	0.0044	0.0002
3:500	0.0284	40.5118	93.05%	98.46%	99.01%	0.9611	0.0054	0.0007

3.1.4. Depth Reconstruction Results of the Proposed Algorithm on Extreme SBR Data

The depth reconstruction results of the proposed algorithm are demonstrated on data with three typical extremely low SBRs of 1:200 and 1:300, as shown in Figure 8.

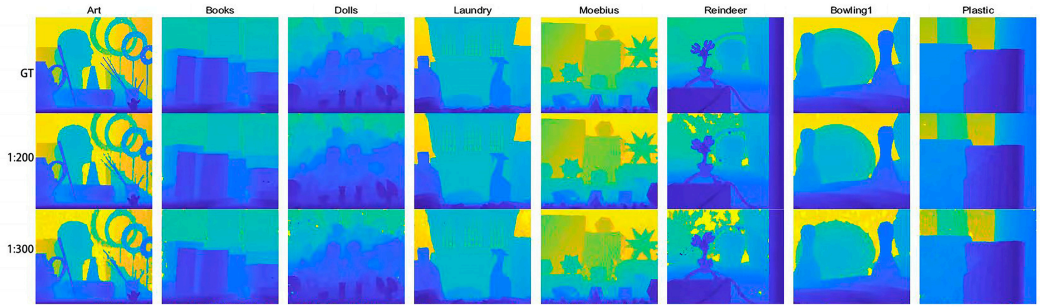


Figure 8. Ground truth and depth reconstruction results on the Middlebury dataset at SBR = 1:200 and 1:300.

3.2. Ablation Analysis of AMAD-Net Components

To effectively evaluate the individual contribution and synergistic effects of each module, ablation experiments were conducted on the network architecture under SBR conditions of

1:100, 2:100, and 3:100. The results are summarized in Table 10. A combinatorial approach was adopted to examine the functions of the sub-modules within AMAD-Net, which include the WAGM, 4DAM, and the Dilated Convolution-based pixel-wise denoising module (DC). In Table 10, a checkmark indicates the inclusion of the corresponding sub-module.

Table 10. Network architecture ablation experiments.

SBR	WAGM	4DAM	DC	RMSE (m)	PSNR (dB)	Accuracy < 1.01	SSIM	AbsRel	SqRel
		✓	✓	0.0334	39.1406	88.82%	0.9519	0.0066	0.0010
	✓		✓	0.0228	41.1331	89.69%	0.9559	0.0059	0.0003
	✓	✓		0.0234	41.0521	90.52%	0.9529	0.0057	0.0004
1:100	✓			0.0483	37.7418	89.53%	0.9481	0.0072	0.0031
		✓		0.0640	36.2670	87.60%	0.9440	0.0089	0.0063
			✓	0.0237	40.7231	91.23%	0.9554	0.0055	0.0003
	✓	✓	✓	0.0219	41.4678	91.39%	0.9556	0.0054	0.0003
		✓	✓	0.1087	32.1779	80.22%	0.9154	0.0154	0.0180
	✓		✓	0.0513	35.6698	83.79%	0.9298	0.0087	0.0023
	✓	✓		0.0462	36.7760	84.47%	0.9232	0.0085	0.0021
1:200	✓			0.1509	27.9902	80.48%	0.8943	0.0199	0.0282
		✓		0.2021	25.7657	78.49%	0.8889	0.0273	0.0519
			✓	0.0652	34.1917	84.36%	0.9220	0.0096	0.0044
	✓	✓	✓	0.0512	36.5166	85.31%	0.9311	0.0083	0.0030
		✓	✓	0.2969	19.8320	65.53%	0.8277	0.0443	0.0652
	✓		✓	0.1401	26.6228	69.85%	0.8313	0.0236	0.0172
	✓	✓		0.1439	26.3869	73.52%	0.8568	0.0212	0.0163
1:300	✓			0.3933	16.9593	68.14%	0.7999	0.0569	0.1082
		✓		0.4865	15.0911	68.25%	0.8003	0.0715	0.1686
			✓	0.2053	23.1911	66.16%	0.8013	0.0356	0.0363
	✓	✓	✓	0.1131	30.6315	77.05%	0.8900	0.0162	0.0173

3.3. Ablation Analysis of Loss Function

To effectively evaluate the role of the loss function, ablation experiments on the loss functions were conducted under the condition of a 1:100 SBR dataset. Based on AMAD-Net, the effects of various loss functions were examined, specifically testing three types of loss functions: CE, CE + TV, and CE + NLTV, as shown in Table 11.

Table 11. Loss Function ablation experiments.

Types of Loss Function	RMSE (m)	PSNR (dB)	Accuracy (δ = 1.01)	SSIM	AbsRel	SqRel
CE	0.0305	38.5743	87.95%	0.9537	0.0066	0.0006
CE + TV	0.0273	40.3221	90.01%	0.9538	0.0060	0.0006
CE + NLTV (Proposed)	0.0219	41.4678	91.39%	0.9556	0.0054	0.0003

4. Discussion

This study proposes an Asymmetric Multi-scale Attention-Guided Denoising Network, which effectively solves the depth reconstruction challenge of long-range single-photon LiDAR under conditions of extremely sparse signals. In long-range detection, limited by laser power attenuation and strong background light interference, the target echo signals exhibit extreme sparsity. Experiments demonstrate that AMAD-Net can achieve high-precision signal photon extraction and depth reconstruction in extremely low SBR environments.

From the perspective of quantitative evaluation results, the performance of AMAD-Net is significantly superior to existing mainstream single-photon depth reconstruction

algorithms. Taking a complex scene with an SBR of 1:100 as an example, the RMSE of the proposed method is reduced to 0.0219 m, while that of the second-best Yao algorithm is 0.0658 m. Meanwhile, it also exhibits significant advantages in SSIM and PSNR. Based on the experimental results on extremely low SBR data, the number of signal photons is a critical factor influencing the depth reconstruction performance. Under the same SBR conditions, a greater number of signal photons yields better reconstruction results. From a visual perspective, the AMAD-Net algorithm is capable of accomplishing depth reconstruction under the extremely low SBR condition of 1:300.

This substantial performance improvement is mainly attributed to the physical and mathematical logic behind the core modules designed in this paper: The WAGM utilizes the transient high-frequency characteristics of signal photons in the time domain and the Poisson distribution characteristics of random noise. By performing high-frequency feature extraction and applying attention masks in the wavelet domain, the network can effectively suppress background noise at the front end while avoiding the signal truncation caused by conventional soft-thresholding denoising.

The 4DAM effectively addresses the feature extraction bottleneck under extremely low SBR. Since signal photons are extremely sparse, extracting features directly in the time domain is highly prone to being misled by random noise peaks. By combining channel, temporal, frequency-domain, and spatial correlations, particularly utilizing the characteristic that signal photon energy is concentrated in low-frequency components in the frequency domain, 4DAM can pinpoint the true signal regions from both macroscopic and microscopic perspectives.

The sawtooth-arranged convolutions introduced in the pixel-wise denoising stage mathematically achieve the expansion of the receptive field without sacrificing spatial resolution. This enables the network to capture the spatial self-similarity of sparse signals within a broad neighborhood while taking into account both high-frequency details and low-frequency information. Furthermore, the introduction of the NLTV loss function overcomes the edge blurring problem caused by the traditional TV loss, retaining sharp edge information while filtering out artifacts.

This depth reconstruction capability under extremely low SBR conditions holds immense potential value for the practical applications of long-range LiDAR. Particularly in critical fields such as aerospace early warning, targets are often extremely distant, possess small radar cross-sections, and move rapidly. This causes the returning signal photons to be extremely weak and submerged in strong background noise. AMAD-Net can still maintain a high reconstruction accuracy under extreme SBR conditions as low as 1:300, providing highly robust technical support for the early detection, precise ranging, and 3D profile recognition of long-range aerial targets.

Objectively speaking, although AMAD-Net performs exceptionally well in long-range weak signal detection, as defined earlier, the current model still has certain limitations. In pursuit of ultimate single-peak localization accuracy, the current classification-based reconstruction strategy simplifies long-range imaging to single-peak scenarios. Consequently, it lacks the capability to handle complex multi-peak scenarios and cannot address the multi-echo phenomena within a single pixel caused by semi-transparent targets or target edge occlusions. Exploring multi-echo feature extraction and reconstruction mechanisms under extremely low SBR conditions will be a key direction for expanding this model in our next stage of research.

Furthermore, to address the bottlenecks of challenging data acquisition and high annotation costs of high-quality depth labels in real-world single-photon active imaging, training networks on simulated datasets followed by cross-domain validation on real-world data has emerged as a widely adopted and effective paradigm in this field. Previous litera-

ture [14–20] has demonstrated that depth reconstruction models trained on the simulated NYUv2 dataset exhibit robust generalization capabilities on real-world scene datasets. Benefiting from the rich geometric topology and diverse spatial contexts of the NYUv2 dataset, the trained models can profoundly learn generalized 3D spatial representations and edge priors. Prior studies indicate that such models, trained solely on simulated NYUv2 data, can still maintain excellent denoising and structural restoration performance when directly transferred to real-world scenes captured by single-photon LiDAR. For instance, after training their network on simulated data, Lindell et al. [14] directly and successfully validated it on real-world datasets captured by the LinoSPAD hardware system. Similarly, Zhao et al. [18] and Zhang et al. [20] applied their simulation-trained models to the NJUST real-world LiDAR dataset, both achieving outstanding depth reconstruction results. These works fully demonstrate the formidable "simulation-to-real" generalization capability of this scheme. This not only proves the efficacy of the physical simulation-based training paradigm but also establishes a solid theoretical and experimental foundation for mitigating the data scarcity issue in real-world depth reconstruction under extremely low signal-to-noise ratios.

5. Conclusions

To address the challenge of extremely low data signal-to-background ratios in long-range single-photon LiDAR depth reconstruction, this paper proposes an Asymmetric Multi-scale Attention-Guided Denoising depth reconstruction network. AMAD-Net suppresses noise photon counts through a wavelet denoising preprocessing module and effectively refines depth features by integrating a four-dimensional attention mechanism. The application of zigzag dilated convolutions in the pixel-wise denoising module effectively enhances the detection performance of weak signal photons. Furthermore, the joint loss function combining cross-entropy and non-local total variation regularization improves the reconstruction performance for data with extremely low SBR. The generalization capability of the depth reconstruction algorithm is validated on the Middlebury dataset. Finally, the effectiveness of each algorithm module and the NLTV loss function is verified through ablation experiments on the network architecture and loss functions.

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